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PHYSICS
Paper 1
July./ Aug.2026
$2\frac{1}{2}$ hours

BOARD OF GOVERNORS EXAMINATION CONSULTS

Uganda Certificate of Education

PHYSICS

Paper 1
(Theory)

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

*This examination paper has **two** Sections; **A** and **B**.*

*It consists of **seven** examination items.*

*Section **A** has **three compulsory** items.*

*Section **B** has **two** parts **I** and **II**. Respond to **only one** item from each part.*

*Respond to **five** examination items in all.*

*Any additional item(s) responded to will **not** be scored.*

***All** responses **must** be written in the answer booklet(s) provided.*

Responses to each item should start on a fresh page.

Indicate the item number you are responding to, clearly at the top of the page.

Graph Paper is provided.

Mathematical tables and silent non-programmable calculators may be used.

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Turn Over

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SECTION A

Respond to **all** *items in this section*

Item 1

During a weekend youth conference at a beach resort along the shores of Lake Victoria, participants saw interesting optical and wave phenomena. In the afternoon, while driving through the access road, some saw what appeared like a pool of water on the dry road ahead that cleared as they went closer to it.

Later that evening, during an indoor gala in the resort room lit with cyan lights, participants wearing magenta outfits were surprised to find that their clothes appeared in a different color and wondered how this was possible.

Before going to sleep, a group of participants tried listening to a live football broadcast from a radio station in Entebbe on a portable radio at a frequency of 96.7MHz, but as they moved deeper into the resort rooms, the signal reduced and finally went off completely. One of them began a debate saying radio waves don't exceed 10m and added that radio waves are electromagnetic waves and could have other uses in our daily life.

Hint: The speed of electromagnetic waves is $3.0 \times 10^{8} ms^{-1}$.

Tasks: They have suggested that you clear the participants debates and surprises like;

(a) Why some of their outfits appeared a different colour under the room lights?
(b) The phenomenon that caused a pool of water like appearance on the hot road ahead.
(c) Explain how the radio waves failed to reach the deeper resort room.
(d) Describe other uses of electromagnetic radiations in their daily-life.

Item 2

During a routine check at a steel manufacturing plant, an engineering team used Cobalt-60 in a sealed radioactive source to inspect welds for hidden cracks in a heavy machinery room.

After completing the inspection, there was an accidental spill that occurred in the room and the Senior Officer immediately sealed it off and instructed any person who wanted to access the room to wait for the right time but he only told them that the right time is when the activity is equal to the background count rate of 40 counts per minute which made the people confused.

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The Senior officer also guided them to observe precautions that he gave them when they enter the room.

While waiting, the people noticed a radiation hazard sign on an automated metal sheet thickness gauge, which the Safety Officer explained that it uses a beta-emitting radioisotope (Strontium-90). The people wondered how such hazardous materials can be safely used in daily-life and why the precautions must be taken. The safety team recorded the decay activity of the spilled substance over time as shown in Table:

Activity (counts per minute)	480	280	163	95	55	32
Time (hours)	0	10	20	30	40	50

Tasks: As a learner of physics help the people to know;

a) How long they have to wait before they are allowed to enter the room? (Hint: Use graphical method to analyze the data in the table)

b) Why they need to wait for that time before entering the isolated room?

c) The precautions that to observe when they enter the room.

d) The ways in which such hazardous materials can be safely used in daily-life.

Item 3

During an interactive video broadcast live on TV presented by a biologist from a research station in Sydney, the students streaming live from Nairobi notice that while they are sitting in broad daylight, the researcher in Sydney is standing under a dark midnight sky with a bright full moon overhead, observing ocean water rising rapidly along the shoreline during high tide.

The students are curious and want to know; why it can be dark midnight in Sydney while it is broad daylight in Nairobi at the exact moment, the live video and audio from Sydney can reach their classroom screen in Kenya almost instantaneously across thousands of kilometers, how the Moon located far away in space, is able to pull earth's ocean water to bring about high and low tides.

Task: Using your knowledge of physics, explain the scientific principles behind each of the students' concerns.

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SECTION B

PART I

Attempt **only one** *item from this part*

Item 4

After descending a steep road for 5 minutes at a constant speed, a fully loaded 100 tonne dump truck driver suddenly spots a stalled-bulldozer blocking the narrow road. The driver immediately applies emergency brakes, and the heavy truck takes 60 seconds to come to a complete stop, narrowly making contact with the bulldozer.

On stopping, a portion of the heavy iron ore cargo shifted violently forward, safety inspectors arrived on scene and record the following observations;

- ✓ The truck was carrying a maximum payload, the brake linings and tyres were heavily worn, and extreme friction heat was emitted from the wheel assemblies.
- ✓ The truck covered a total distance of 4.95 km from the moment it began moving at constant speed until it uniformly decelerated to rest.
- ✓ The truck utilizes heavy alloy steel brake discs with a mass of 30 kg per wheel assembly, and approximately 9,450 kJ of thermal energy was generated at each wheel during the emergency deceleration.

Hints:

- The mine site insurance policy covers equipment damage only if the vehicle was operated within the safe road speed limit of 50km/h.
- Specific heat capacity of alloy steel brake discs of $450 Jkg^{-1}K^{-1}$.
- Initial disc temperature before braking was 25°.
- Maximum efficient brake operating temperature range is 550°C to 680°C.

Tasks: As a learner of physics, prepare a report to help the management to;

(a) Determine whether the insurance company will compensate for the collision damage.

(b) Explain the physical reasons why the loaded dump truck required a long stopping time and generated intense heat during braking.

(c) Evaluate whether the brake discs remained suitable and effective for safe operation during the incident.

(d) Explain how friction operates to decelerate and stop heavy machinery.

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Item 5

Maya works as a wildlife ranger who walks across a 12km lake daily to reach a remote station. One morning, she set off at exactly 7:30am in her motorboat, carrying warm soup in a single-walled aluminum flask. As she navigated across the lake, she noticed a continuous grinding and high heat coming from the boat's propeller shaft bearings. The sound made her journey uncomfortable, and she wondered what was causing the noise and heat.

After traveling 8km, Maya checked her watch and realized it was already 8:10am, meaning she was likely to be late for her morning briefing. To save time, she opened the throttle to increase her speed. The boat's velocity increased from $3ms^{-1}$ to $11ms^{-1}$ over a period of 4 seconds, which enabled her to arrive at the docking station on time. Later, her teammate asked her about the boat's acceleration during that speed boost, but she could not work it out.

During her lunch break, Maya opened her flask expecting to enjoy hot soup, but found it was only warm. Disappointed, she shared this with a fellow ranger, who advised her to use a vacuum flask because it keeps liquids hot for much longer. Maya became curious about how a vacuum flask works to reduce heat loss and where else the processes of heat loss are applied in everyday life.

Task: Maya has approached you for assistance to;

(a) Explain the likely cause of the noise and heat from the propeller shaft and suggest how it can be minimized.

(b) Determine the boat's acceleration during the journey.

(c) Explain how a vacuum flask keeps liquids hot.

(d) Describe an everyday or industrial application of the heat transfer process that led to the cooling of the soup.

PART II

Attempt **only one** *item from this part*

Item 6

During a national science and innovation exhibition, a team of learners constructed a working model demonstrating how electrical energy generated at a hydroelectric dam is elevated to high voltages, transmitted over long distances and reduced at substations before reaching local communities.

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Meanwhile, the management of their school is reviewing its monthly operational expenses. They rely on several key electrical appliances daily as listed in the table below.

Appliance	Quantity	Rating	Time used daily (hours)
Outdoor security LED lamp	6	240V, 10W	12
Medical autoclave sterilizer	1	240V, 1500W	4
Vaccine refrigerator	1	240V, 200W	24

The health centre management has set aside a monthly budget of **UGX 300,000** for electricity bills (calculated for a standard 30-day month), but the administrator is concerned about whether their appliance usage will remain within this budget. The cost of electricity is UGX.700 per unit (kWh).

Task: As a physics student, assist the school management and team of learners by addressing the following;

(a) Explain how electrical power is transmitted from the distant generation station to the health centre with minimal energy loss.

(b) Justify why parallel connections are recommended for domestic and commercial appliances in the health centre.

(c) Calculate the clinic's monthly energy consumption and determine whether their expected electricity bill is within their allocated budget.

(d) Explain the principle and application of a step-down transformer used in running low-voltage diagnostic equipment and chargers from the mains supply.

Item 7

During an intense thunderstorm, lightning flashed directly above the local community radio station. A few minutes later, smoke began coming from the main audio transmitter console inside the studio, triggering a small electrical fire. After the fire was put out, an electrical safety inspector discovered that the station's roof-top metal antenna mast lacked proper earthing/ grounding. He suspected that lightning had struck the mast, causing a massive surge of current to travel through the feed cable directly into the transmitter.

The inspector advised the station manager to install a lightning arrestor on the building, fit an automatic circuit breaker in the main supply switch-board, and insert the correctly rated fuse inside the transmitter unit. However, the non-technical station manager was

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confused by these terms and did not understand how each device functions to prevent such disasters.

Hints

- The main audio transmitter unit is rated 440 W, 220 V.
- Standard fuse ratings available on the market are 1A, 2A, 3A, 5A, and 13A.

Task: Assist the radio station manager to;

(a) Explain how a lightning arrestor protects the building and equipment from lightning strikes.

(b) Calculate the operating current of the audio transmitter and select the correct fuse rating to be installed.

(c) Explain how an automatic circuit breaker operates to protect electrical appliances against excessive current.

END

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